RATIONAL MAPS TO A LOCALLY FINITELY PRESENTED SCHEME ARE DETERMINED BY THEIR VALUES AT THE GENERIC POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The second paragraph of the proof concludes by citing parts (1) and (2) of an in-chapter lemma, but the reduction to the irreducible case in the first paragraph is the author's own work inside the counted span (separating the components of the source and factoring the correspondence as a product over them), so this is not a citation-only span of the kind refused for two candidates in the sibling half of this chapter.

Allowed standard prerequisites (editorial).

- A morphism is determined by, and locally of finite presentation morphisms are exactly recovered from, the local maps at the generic point. Lemma \ref{lemma-morphism-defined-local-ring}, cited at proof lines 12655-12657. Supplies parts (1), (2) and (3) for an irreducible source.
- The construction of the map from a rational map by evaluation and pullback of local rings. the definition of a rational map and its local rings at the generic points, used at proof lines 12641-12644. Makes the correspondence of the statement well defined.

Source reference. Original lemma 0BX8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14971–14971. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0BX6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14969–14969. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let S be a scheme. Let X and Y be schemes over S. Assume X has finitely many irreducible components with generic points $x_1, \ldots, x_n$. Let $s_i \in S$ be the image of x_i. Consider the map*

$$\left\{\begin{array}{c} S\text{-rational maps} \\ \text{from } X \text{ to } Y \end{array}\right\} \longrightarrow \left\{\begin{array}{c} (y_1, \varphi_1, \ldots, y_n, \varphi_n) \text{ where } y_i \in Y \text{ lies over } s_i \text{ and} \\ \varphi_i : \mathcal{O}_{Y,y_i} \to \mathcal{O}_{X,x_i} \text{ is a local } \mathcal{O}_{S,s_i}\text{-algebra map} \end{array}\right\}$$

which sends $f : U \to Y$ to the $2n$-tuple with $y_i = f(x_i)$ and $\varphi_i = f^\sharp_{x_i}$. Then

1. *If $Y \to S$ is locally of finite type, then the map is injective.*
2. *If $Y \to S$ is locally of finite presentation, then the map is bijective.*
3. *If $Y \to S$ is locally of finite type and X reduced, then the map is bijective.*

Proof. Observe that any dense open of X contains the points x_i so the construction makes sense. To prove (1) or (2) we may replace X by any dense open. Thus if $Z_1, \ldots, Z_n$ are the irreducible components of X, then we may replace X by $X \setminus \bigcup_{i \neq j} Z_i \cap Z_j$. After doing this X is the disjoint union of its irreducible components (viewed as open and closed subschemes). Then both the right hand side and the left hand side of the arrow are products over the irreducible components and we reduce to the case where X is irreducible.

Assume X is irreducible with generic point x lying over $s \in S$. Part (1) follows from part (1) of Lemma 0BX6. Parts (2) and (3) follow from part (2) of the same lemma. □